

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 Correlation and Pattern Analysis

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO34– Correlation and Pattern Analysis	Assessment:	ASSESSMENT TOOL 3 Correlation and Pattern Analysis
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and interpret correlations and hidden patterns in datasets using statistical techniques to identify confounding factors and support reliable decision-making. (BL4)		
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Section / Batch:	2024	Date:	21.08.2026

Code of Conduct

I Jeya Harshini R (192424051) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jeya Harshini R

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding ; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Question 4

Social Media Usage and Academic Performance

Problem Statement

A university collects data from students and discovers that students who spend more time on social media have lower average marks. The correlation coefficient between social media usage and academic performance is negative. Based on this observation, the university proposes restricting social-media access for students.

Before implementing the policy, an analyst discovers the following:

- Students with heavy social-media usage also tend to sleep fewer hours.
- Some students use social media primarily for academic collaboration.
- First-year students have both higher social-media usage and lower average marks than final-year students.
- Students studying certain courses use online platforms extensively.
- The dataset contains a small group of extremely high social-media users.

The task is to determine whether the observed relationship truly indicates that social media causes lower academic performance.

1. Identify the Core Analytical Issue

The central issue is the difference between **correlation and causation**.

The university observes a negative correlation between social-media usage and academic performance and immediately assumes that social media is the cause of poor academic performance.

However, correlation only shows an association between two variables and does not establish a direct causal relationship.

Therefore, the university's conclusion may be misleading.

2. Interpretation of the Observed Correlation

The negative correlation indicates that students who spend more time on social media generally tend to have lower marks.

However, this relationship does not necessarily mean that social media itself is harmful.

Several alternative explanations may exist:

- Social media may reduce study time.
- Poor-performing students may spend more time on social media.
- Other variables may influence both social-media usage and academic performance.

Therefore, the correlation must be interpreted carefully.

3. Potential Confounding Variables

Sleep Duration

The analyst found that heavy social-media users sleep fewer hours.

Sleep deprivation is known to affect:

- Memory
- Concentration
- Learning ability
- Academic performance

Thus, poor marks may be caused by insufficient sleep rather than social-media usage itself.

Academic Collaboration

Some students use social-media platforms for:

- Group discussions
- Sharing notes
- Project collaboration
- Academic support

In such cases, social-media usage may actually improve academic performance.

Combining these students with recreational users can distort the analysis.

Year of Study

First-year students:

- Use social media more frequently.
- Generally have lower academic performance due to adjustment challenges.

Thus, year of study may influence both variables simultaneously.

Course Type

Different academic programs require different levels of online interaction.

For example:

- Computer Science students may use Discord, GitHub, Reddit, and LinkedIn extensively.
- Media students may spend substantial time on social platforms for coursework.

Hence, course type may affect social-media usage without necessarily reducing academic performance.

4. Possible Subgroup Effects

The relationship may vary across different groups.

First-Year Students

May show a strong negative relationship.

Final-Year Students

May show little or no relationship.

Technical Courses

May use social media productively.

Non-Technical Courses

May exhibit different patterns.

Combining all students into one dataset can hide important subgroup differences.

5. Outlier Effects

The dataset contains a small group of extremely heavy social-media users.

These students may:

- Spend unusually large amounts of time online.
- Have exceptionally low marks.

Such extreme observations can significantly influence the correlation coefficient.

Therefore:

- Boxplots should be created.
- Outliers should be examined separately.
- Sensitivity analysis should be performed.

The correlation should be recalculated with and without outliers.

6. Alternative Explanations

Several alternative explanations exist:

Reverse Causality

Students with poor academic performance may spend more time on social media because they are less engaged with studies.

Time Management Problems

Students with poor time-management skills may both:

- Use social media excessively.
- Perform poorly academically.

Motivation Levels

Less motivated students may show both higher social-media usage and lower academic achievement.

Stress and Mental Health

Students experiencing stress may spend more time online and simultaneously struggle academically.

7. Critical Evaluation of the University's Conclusion

The university proposes:

"Restrict social-media access for students."

This conclusion is premature because:

1. Correlation does not imply causation.
2. Several confounding variables exist.
3. Different student groups may exhibit different relationships.
4. Outliers may distort the overall pattern.
5. Social media may sometimes support learning.

Therefore, implementing a restrictive policy based solely on correlation would be inappropriate.

8. Recommended Analytical Strategy

Visualization

Create:

- Scatter plots
- Histograms
- Boxplots

to observe patterns and outliers.

Subgroup Analysis

Analyze separately by:

- Year of study
 - Department
 - Course type
-

Correlation Analysis Within Groups

Calculate correlations for each subgroup.

This helps determine whether the relationship remains consistent.

Multiple Regression Analysis

Use:

$$Marks = \beta_0 + \beta_1(SocialMedia) + \beta_2(Sleep) + \beta_3(Year) + \beta_4(Course)$$

This allows the independent effect of social-media usage to be estimated.

Outlier Investigation

Recalculate results after removing extreme observations.

Compare outcomes.

Statistical Significance Testing

Determine whether social-media usage remains significant after controlling for confounding variables.

Yes — and that's exactly what I'd be worried about if this is for a **Query Processing / Correlation & Pattern Analysis assignment**.

The answers I gave are **theory-heavy**. They satisfy the rubric, but they don't feel very *practical* or *analytical*. Faculty usually like seeing actual analysis steps, tables, plots, and what you would do with the data.

For maximum marks, add a **Practical Analysis Section** to each answer.

For Question 4 (Social Media vs Marks)

Practical Analysis Approach

Step 1: Visualize the Data

Create a scatter plot:

X-axis: Hours spent on social media

Y-axis: Student marks

Purpose:

- Observe overall trend
- Detect clusters
- Identify outliers

Example:

Social Media Hours Marks

2	85
3	80
5	70

Social Media Hours Marks

7	60
10	40

Expected pattern:

- Marks generally decrease as usage increases.

Step 2: Detect Outliers

Use:

- Box Plot
- Z-Score Method
- IQR Method

Example outlier:

Student Hours Marks

A	15	10
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This student may heavily influence the correlation coefficient.

Recalculate correlation after removing outliers.

Step 3: Subgroup Analysis

Divide students into groups:

Group Social Media Use

Low	< 2 hours
Medium	2–5 hours
High	> 5 hours

Compare average marks in each group.

Also analyze:

- First Year
- Second Year
- Third Year

- Final Year

separately.

Step 4: Correlation Matrix

Calculate correlations among:

Variable	Marks
----------	-------

Social Media Usage ?	
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Sleep Hours	?
-------------	---

Study Hours	?
-------------	---

Attendance	?
------------	---

This helps identify hidden factors.

Step 5: Multiple Regression

Model:

$$Marks = \beta_0 + \beta_1(SocialMedia) + \beta_2(Sleep) + \beta_3(StudyHours) + \beta_4(Attendance)$$

Interpretation:

If Social Media becomes insignificant after adding Sleep Hours, then sleep may be the true cause.

Step 6: Final Decision

Only after controlling for:

- Sleep
- Study Hours
- Attendance
- Academic Year

can we determine whether social media truly affects marks.

Conclusion

The observed negative correlation does not provide sufficient evidence that social media directly causes lower academic performance.

Several confounding variables—including sleep duration, academic collaboration, year of study, and course type—may influence the relationship.

A detailed analysis involving subgroup analysis, regression modeling, outlier investigation, and significance testing should be performed before implementing any university-wide policy.

Question 5

Advertising Expenditure and Sales Growth

Problem Statement

A retail company analyzes five years of monthly sales data and discovers a very strong positive correlation:

$$r = 0.93$$

between advertising expenditure and sales revenue.

Management concludes:

"Advertising is responsible for the increase in sales, so increasing the advertising budget will proportionally increase revenue."

However, additional investigation reveals:

- New stores were opened.
 - Product prices increased.
 - Customer base expanded.
 - Seasonal sales peaks occur every year.
 - Online sales were introduced during the third year.
 - Advertising expenditure increased steadily over time regardless of short-term sales performance.
-

1. Identify the Core Analytical Issue

The primary issue is the possibility of a **spurious correlation** caused by time-dependent trends.

A strong correlation alone cannot establish causality.

2. Interpretation of the Correlation

A correlation coefficient of:

$$r = 0.93$$

indicates a very strong positive relationship.

However, it only shows that advertising expenditure and sales moved together.

It does not prove that advertising caused sales growth.

3. Why the Correlation May Be Misleading

Common Time Trend

Both variables increased over five years.

When two variables rise over time, a high correlation can occur even without a causal relationship.

Store Expansion

New stores increase:

- Sales volume
- Customer reach

independent of advertising.

Price Increases

Revenue may rise because products become more expensive.

Higher revenue does not necessarily indicate higher demand.

Growth in Customer Base

More customers naturally lead to more sales.

Online Sales Introduction

The introduction of e-commerce can create a major increase in sales independent of advertising.

Seasonal Patterns

Holiday seasons often create recurring sales peaks.

Advertising and sales may both increase during these periods.

4. Critical Evaluation of Management's Conclusion

Management assumes:

More advertising = proportionally more sales.

This conclusion is not supported because:

- Other factors changed simultaneously.
- Trend effects may inflate correlation.
- Seasonal effects may distort relationships.
- Store expansion and online sales may explain growth.

Therefore, the evidence is insufficient to establish causality.

5. Patterns That Must Be Investigated Separately

Long-Term Trend

Determine whether sales continue increasing after removing trend effects.

Seasonality

Examine:

- Monthly patterns
 - Holiday peaks
 - Festival periods
-

Store-Level Performance

Compare:

- Existing stores
 - Newly opened stores
-

Online vs Offline Sales

Analyze separately.

Customer Growth

Examine sales per customer rather than total sales.

Pricing Effects

Adjust revenue for price changes.

6. Recommended Analytical Workflow

Step 1: Visualization

Create:

- Time-series plots
- Scatter plots
- Trend charts

to identify patterns.

Step 2: Time-Series Decomposition

Separate:

- Trend
 - Seasonality
 - Random variation
-

Step 3: Detrending

Remove long-term growth trends.

Then calculate correlations again.

Step 4: Subgroup Analysis

Analyze:

- Online sales
- Offline sales
- Regions
- Product categories

separately.

Step 5: Multiple Regression

Use:

$$\text{Sales} = \beta_0 + \beta_1(\text{Advertising}) + \beta_2(\text{StoreCount}) + \beta_3(\text{Price}) + \beta_4(\text{CustomerBase}) + \beta_5(\text{OnlineSales})$$

This isolates the independent contribution of advertising.

Step 6: Statistical Testing

Perform:

- Significance tests
- Regression coefficient tests
- Model diagnostics

to determine whether advertising remains an important predictor.

Step 7: Lag Analysis

Advertising often influences future sales rather than immediate sales.

Examine:

- One-month lag
- Two-month lag
- Three-month lag

relationships.

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For maximum marks, add a **Practical Analysis Section** to each answer.

For Question 5 (Advertising vs Sales)

This one has even stronger practical content.

Practical Analysis Approach

Step 1: Plot Time-Series Graphs

Graph 1:

Month Sales

Jan 50,000

Feb 55,000

Mar 60,000

Graph 2:

Month Advertising

Jan 5,000

Feb 6,000

Mar 7,000

Purpose:

Observe whether both variables simply increase over time.

Step 2: Scatter Plot

Plot:

X-axis: Advertising Spend

Y-axis: Sales

Check whether relationship is:

- Linear
 - Curved
 - Clustered
-

Step 3: Trend Analysis

Create moving averages.

Example:

$$\text{Moving Average} = \frac{\text{Sales}_1 + \text{Sales}_2 + \text{Sales}_3}{3}$$

Purpose:

Remove short-term fluctuations.

Step 4: Seasonal Analysis

Compare:

Month	Avg Sales
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January	50,000
---------	--------

February	48,000
----------	--------

December	90,000
----------	--------

This identifies festival effects.

Step 5: Before vs After Online Sales

Separate:

Period	Avg Sales
--------	-----------

Before Online Store	60,000
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After Online Store	95,000
--------------------	--------

Check whether sales increased because of online expansion rather than advertising.

Step 6: Store-Level Analysis

Compare:

Store Type	Sales Growth
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Existing Stores	10%
-----------------	-----

New Stores	45%
------------	-----

This identifies the impact of expansion.

Step 7: Multiple Regression Model

$Sales = \beta_0 + \beta_1(Advertising) + \beta_2(StoreCount) + \beta_3(CustomerBase) + \beta_4(Price) + \beta_5(OnlineSales)$

Purpose:

Measure the actual contribution of advertising after controlling for other factors.

Step 8: Hypothesis Testing

$$H_0 : \beta_1 = 0$$

Advertising has no significant effect.

$$H_1 : \beta_1 \neq 0$$

Advertising significantly affects sales.

If:

$$p < 0.05$$

then advertising is statistically significant.

Conclusion

Although advertising expenditure and sales show a strong correlation of 0.93, this does not prove that advertising caused the increase in sales.

Store expansion, price increases, customer growth, seasonality, online sales introduction, and long-term trends may all contribute to the observed relationship.

Only after conducting time-series analysis, subgroup analysis, detrending, regression modeling, lag analysis, and statistical testing can the company determine whether advertising genuinely drives sales growth.